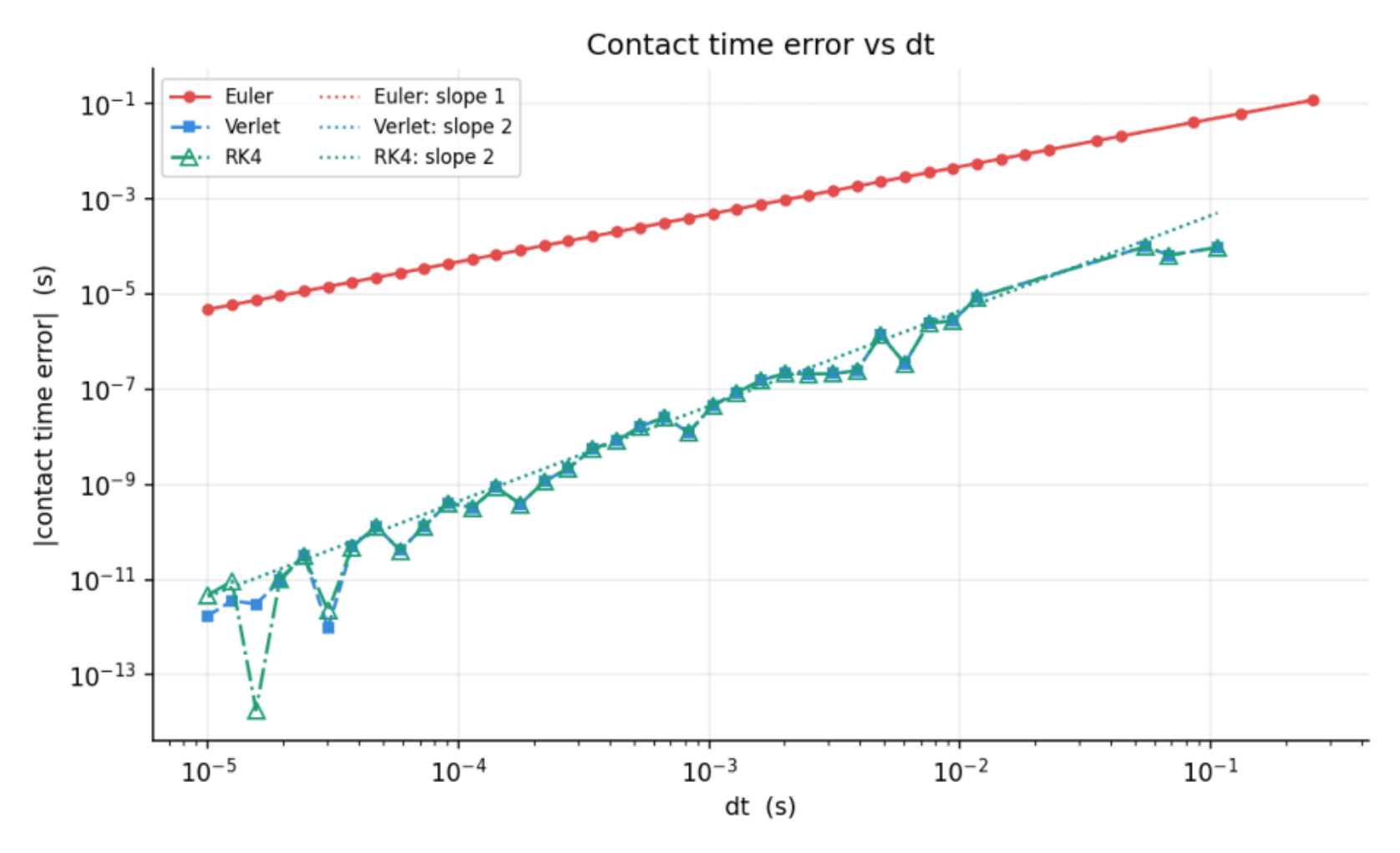
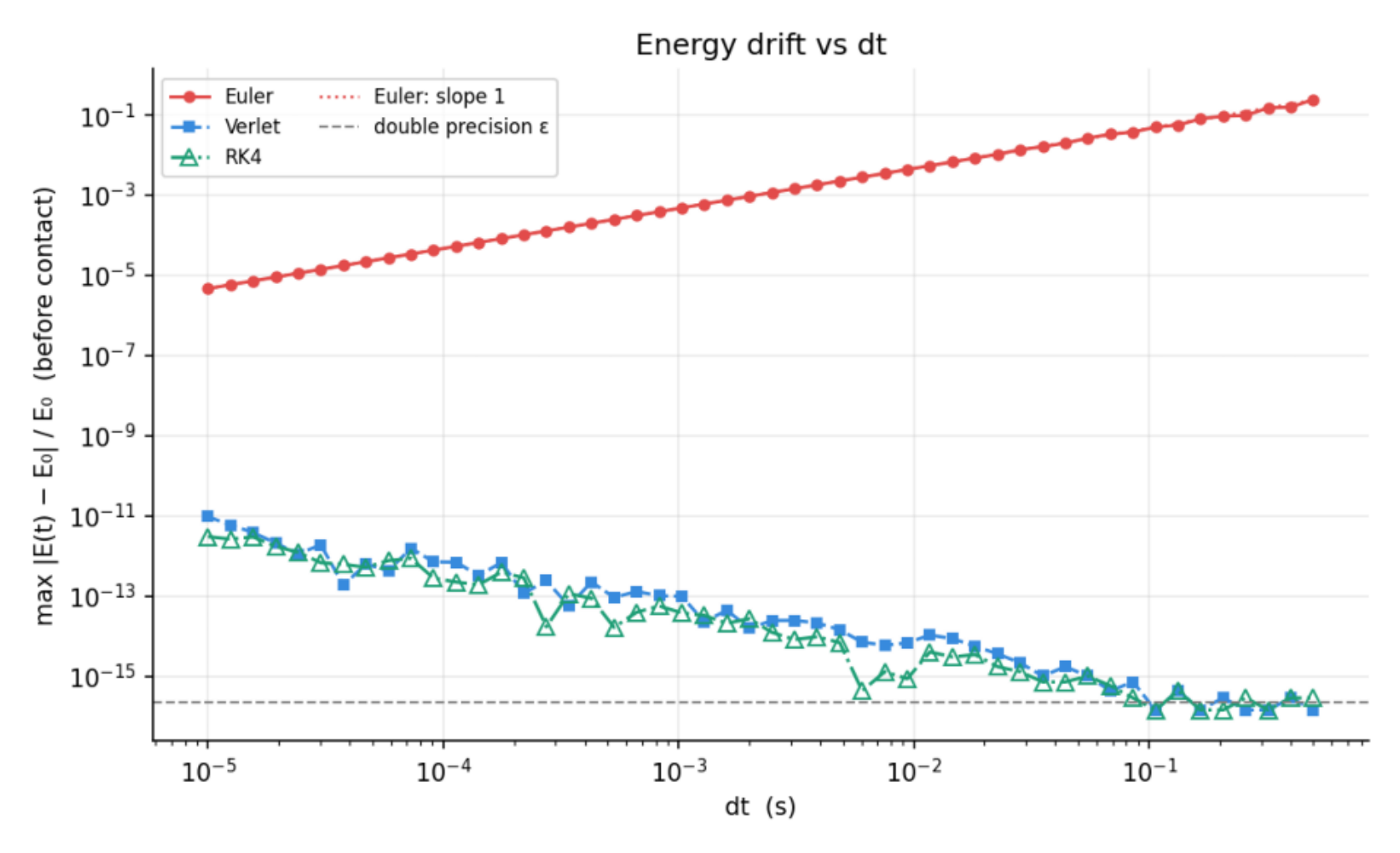
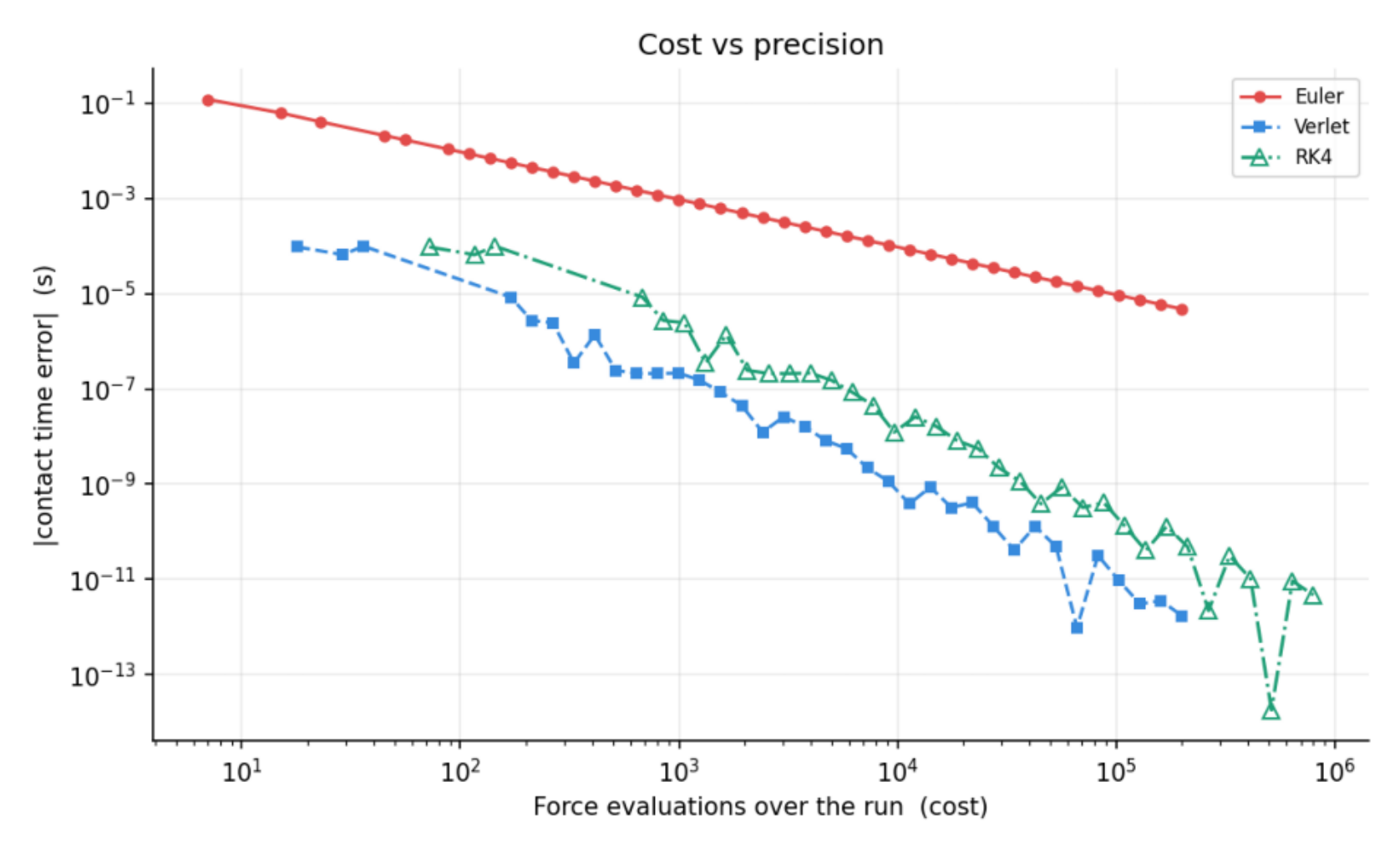
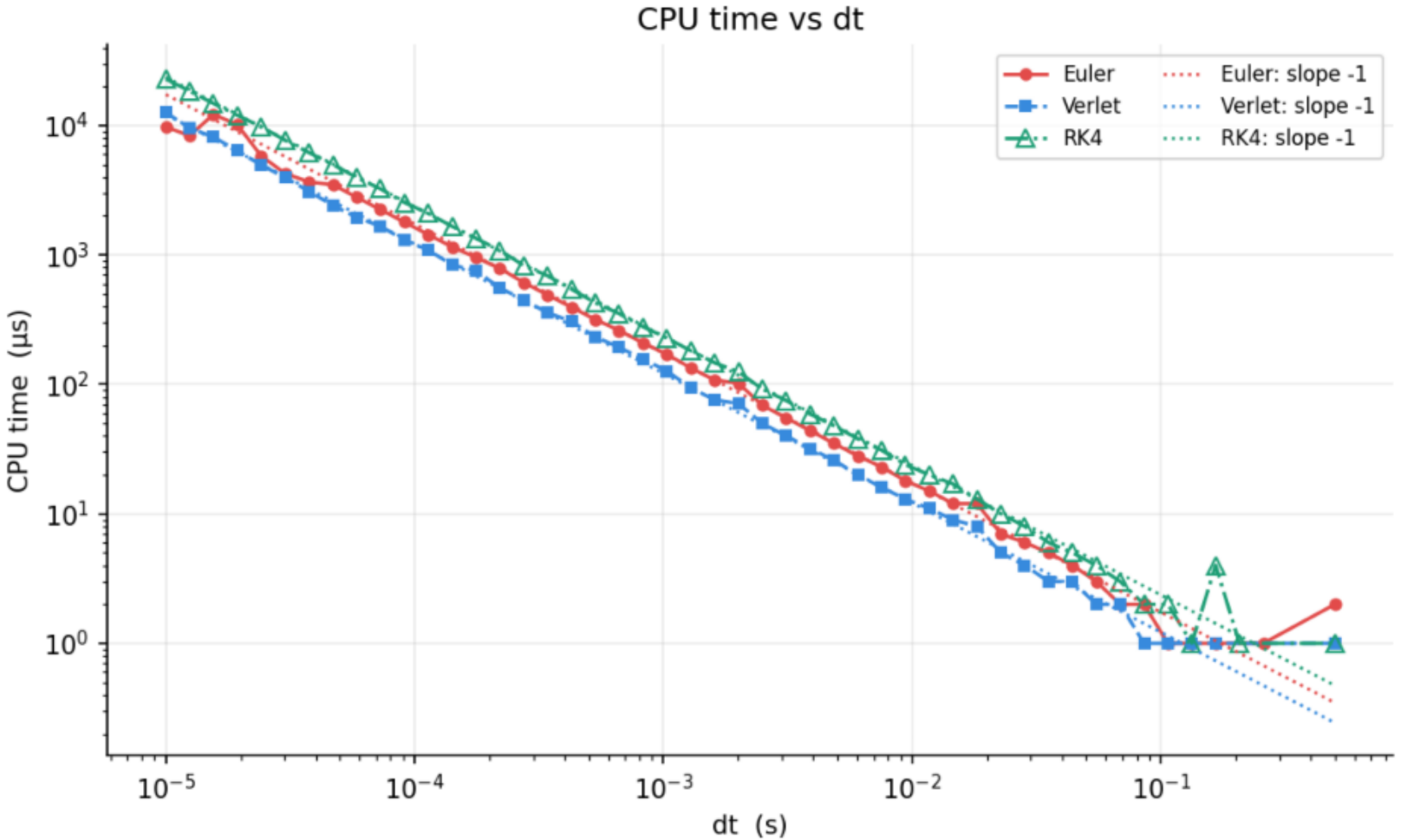
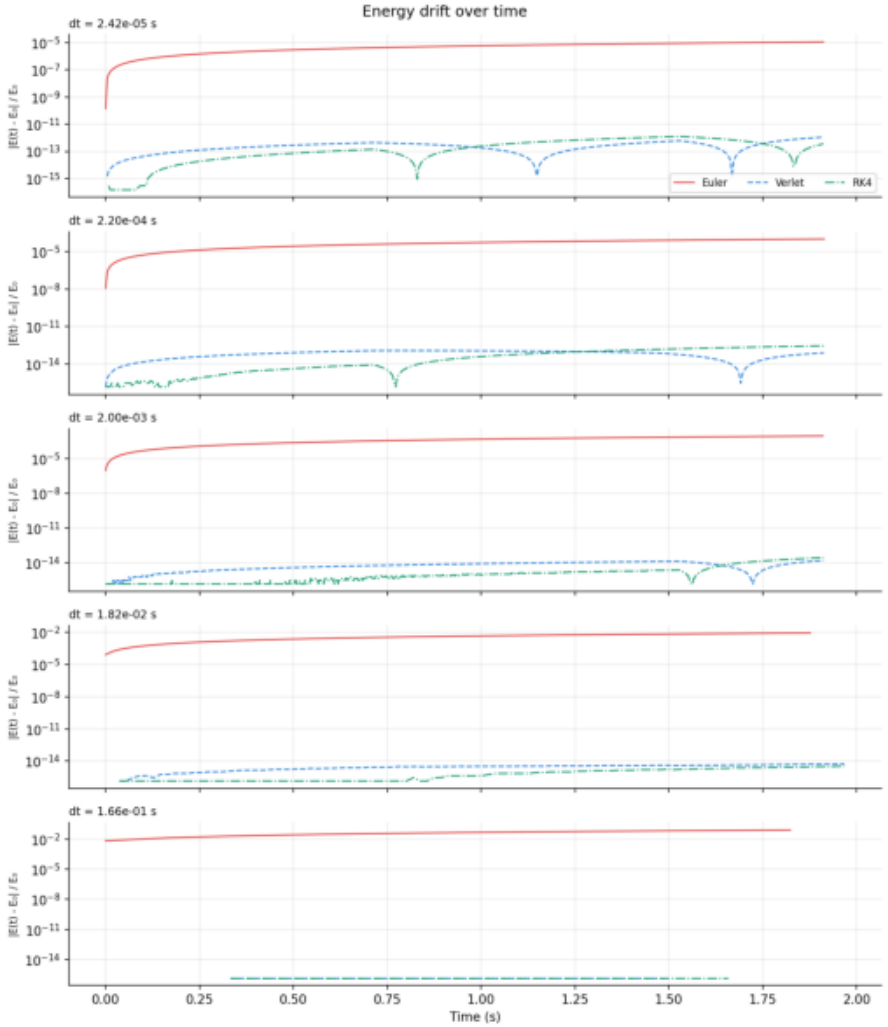

Contact time error vs dt
Euler
Verlet
RK4
Euler: slope 1
Verlet: slope 2
RK4: slope 2
|contact time error| (s)
dt (s)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}


Energy drift vs dt
Euler
Verlet
RK4
Euler: slope 1
double precision ε
max |E(t) − E₀| / E₀ (before contact)
dt (s)
10⁻¹
10⁻³
10⁻⁵
10⁻⁷
10⁻⁹
10⁻¹¹
10⁻¹³
10⁻¹⁵
10⁻⁵
10⁻⁴
10⁻³
10⁻²
10⁻¹


Cost vs precision
Euler
Verlet
RK4
|contact time error| (s)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
10^{6}
Force evaluations over the run (cost)


CPU time vs dt
Euler
Verlet
RK4
Euler: slope -1
Verlet: slope -1
RK4: slope -1
CPU time (µs)
dt (s)
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}


Energy drift over time
dt = 2.42e-05 s
dt = 2.20e-04 s
dt = 2.00e-03 s
dt = 1.82e-02 s
dt = 1.66e-01 s
|E(t) - E₀| / E₀
Euler
Verlet
RK4
Time (s)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00